

TechCorp API

Technical Handbook v1.5

For Developers and Integration Engineers | API Version: v1 | Last Updated: Feb 2025

1. Authentication

1.1 API Keys

All API requests must include an API key in the Authorization header. Keys are generated from Settings > API Keys in the dashboard. Each key is scoped to specific permissions (read, write, admin). Keys should be rotated every 90 days.

Example request header: Authorization: Bearer tc_live_XXXXXXXXXXXXXXXXXXXX

1.2 OAuth 2.0

Enterprise customers can use OAuth 2.0 for user-level authentication. Supported grant types: authorization_code (for web apps) and client_credentials (for server-to-server). Token expiry is 3600 seconds; refresh tokens are valid for 30 days.

1.3 JWT Tokens

All user sessions issue JWTs signed with RS256. The public key is available at <https://api.techcorp.com/.well-known/jwks.json>. Tokens contain: user_id, email, org_id, role, and iat/exp fields. Do not store JWTs in localStorage - use HttpOnly cookies.

2. API Endpoints

Method	Endpoint	Description	Auth Required
GET	/api/v1/health	Service health check	No
POST	/api/v1/auth/token	Issue access token	API Key
GET	/api/v1/users/me	Get current user profile	Bearer
GET	/api/v1/dashboards	List all dashboards	Bearer
POST	/api/v1/dashboards	Create new dashboard	Bearer
GET	/api/v1/dashboards/{id}	Get dashboard by ID	Bearer
PUT	/api/v1/dashboards/{id}	Update dashboard	Bearer
DELETE	/api/v1/dashboards/{id}	Delete dashboard	Bearer (Admin)
POST	/api/v1/documents/ingest	Upload and index document	Bearer
GET	/api/v1/documents	List indexed documents	Bearer
DELETE	/api/v1/documents/{id}	Remove document	Bearer (Admin)
POST	/api/v1/query	Natural language query	Bearer
GET	/api/v1/workflows	List workflows	Bearer
POST	/api/v1/workflows/{id}/trigger	Manually trigger workflow	Bearer

GET	/api/v1/audit-logs	Get audit log entries	Bearer (Admin)
POST	/api/v1/webhooks	Register webhook endpoint	Bearer

3. Request and Response Format

3.1 Request Format

All POST/PUT requests must use Content-Type: application/json. File uploads use multipart/form-data. Rate limits: Starter: 100 req/min, Business: 500 req/min, Enterprise: 2000 req/min. Requests exceeding limits return HTTP 429.

3.2 Response Format

All responses follow this structure: { "success": true, "data": { ... }, "meta": { "page": 1, "total": 42, "request_id": "req_abc123" }, "error": null }. On error, success is false and error contains code and message fields.

3.3 Pagination

List endpoints support cursor-based pagination. Use ?cursor=&limit;=20. Max limit is 100. The meta field contains next_cursor and has_more fields. Do not cache cursors for more than 5 minutes as they may expire.

4. Error Codes

HTTP Code	Error Code	Meaning	Resolution
400	INVALID_REQUEST	Malformed JSON or missing required fields	Check request body schema
401	UNAUTHORIZED	Missing or invalid API key / token	Re-authenticate and retry
403	FORBIDDEN	Insufficient permissions for this resource	Check user role and scope
404	NOT_FOUND	Resource does not exist	Verify resource ID
409	CONFLICT	Duplicate resource creation attempt	Check if resource already exists
422	VALIDATION_ERROR	Field-level validation failed	See errors[] array in response
429	RATE_LIMITED	Request rate exceeded	Back off; retry after Retry-After header value
500	INTERNAL_ERROR	Server-side error	Retry with exponential backoff; contact support
503	SERVICE_UNAVAILABLE	Planned maintenance or overload	Check status.techcorp.com

5. Webhooks

Register a webhook URL to receive real-time event notifications. TechCorp signs all webhook payloads with HMAC-SHA256 using your webhook secret. Verify the X-TechCorp-Signature header before processing any payload.

- dashboard.created / dashboard.updated / dashboard.deleted
- document.ingested / document.deleted
- workflow.triggered / workflow.completed / workflow.failed
- user.invited / user.deactivated

Webhook delivery timeout is 10 seconds. Failed deliveries are retried with exponential backoff (1m, 5m, 30m, 2h, 8h). After 5 consecutive failures the webhook is automatically disabled and an alert email is sent.

6. Troubleshooting Guide

6.1 Token Expired Errors

If you receive 401 UNAUTHORIZED with message 'Token expired', your access token has expired. Use your refresh token to obtain a new access token via POST `/api/v1/auth/refresh`. If the refresh token is also expired (after 30 days), the user must re-authenticate from the login page.

6.2 Slow Query Performance

Queries taking more than 5 seconds are likely hitting large datasets without filters. Add date range filters, use indexed fields for filtering, and enable query caching in dashboard settings. Contact support if p95 latency remains above 3 seconds after optimization.

6.3 Document Ingestion Failures

Common causes: file size above 50MB (compress before upload), unsupported file format (only PDF, DOCX, XLSX, TXT supported), corrupted file, or password-protected PDF. Check the ingestion job status via GET `/api/v1/documents/{id}/status`. Failed jobs return a detailed `error_reason` field.

6.4 Webhook Signature Mismatch

If HMAC verification fails: (1) Ensure you're using the raw request body, not a parsed version. (2) The secret is the webhook-specific secret, not your API key. (3) Use constant-time comparison to prevent timing attacks. Reference implementation available at docs.techcorp.com/webhooks.